SEN Support Platform: Pre-Defense 1 Readiness Review

Repository: github.com/AknurOE/sen-support-platform
Reviewed version: main branch, commit aed3db2 (first review at 8bf7b30), 23 October 2026
Basis: the department document for 24 October (Git repository requirements, minimum requirements and the 11-component evaluation table)
Review date: 23 October 2026. Updated the same day after PR #12 was merged and the supervisor's review was received; the supervisor's feedback is summarised in section 7.

1. Overview

The requirements for this pre-defense fall into two groups. The first is the presentation, which must cover the 11 components listed in the evaluation table. The second is the GitHub repository, which must show that the team has actually been working on the project rather than only preparing slides. This review checks both.

In terms of content, the project is ready. Every one of the 11 components is present in the repository and in the presentation, and most are more detailed than this stage requires. The strongest parts are the link between the user survey, the requirements, the access matrix and the verification plan, the honest separation of verified findings from assumptions, and a working prototype available in Kazakh, Russian and English. The supervisor reviewed the materials on 23 October 2026 and recommended the project for Pre-Defense 1.

The repository side is weaker. There is no pre-defense-1 tag, the supervisor's feedback still has to be merged into the repository log, and YouTrack cannot be opened without a login. There are also smaller process gaps in reviews and commit messages. Most of these can be fixed in a single evening, and section 5 lists the order in which to do it.

Area	Result
Presentation: 11 components	9 done, 2 partial, 0 not done
GitHub: 9 requirements	3 met, 5 partially met, 1 not met
Build and CI	All checks pass
Supervisor review	Received 23 October 2026: 12 comments, all accepted; recommended for Pre-Defense 1

2. Part A. Presentation: the 11 components

A1. Project proposal. Status: done

The project proposal states the problem, the aim, six objectives, the target users and stakeholders, the scope with clear exclusions, the expected outcomes and a section on

engineering complexity. The team and supervisor are named. The scope is realistic for a diploma project because the first implementation milestone is limited to access control and profiles, and the complexity comes from multiple roles sharing child data with different permissions. The supervisor confirmed that the scope is realistic (SF-01).

A2. Evidence of relevance. Status: partial

The team ran an anonymous survey with three parents and four specialists on 22 and 23 October. The survey method and the findings mapped to requirements record 16 findings and mark which are confirmed by at least two respondents and which remain assumptions. Raw answers are not committed because one response contains contact details, and the documents explain this. The gap is that the table asks for three to five discussions or interviews with user representatives, and an anonymous form is not the same thing. The supervisor made the same point (SF-02), and the team has planned three to five semi-structured interviews before Pre-Defense 2.

A3. Literature review. Status: done

The literature review uses 14 peer reviewed publications, 13 of them from 2021 to 2025, with one foundational review from 2019. It includes a structured analysis table and a section that traces each design decision back to the sources. This meets the requirement of at least 12 sources with at least 8 scientific publications. All DOIs resolve. The authors state that the review is focused rather than systematic, which is the correct framing. The supervisor asked for two or three additional sources on access control and a single citation style (SF-03).

A4. Existing system comparison. Status: done

The comparison and its criteria cover four systems (Birdhouse for Autism, Autism Tracker Pro, CareClinic and MyChart Proxy) against ten criteria, with dated public sources. This is within the required range of three to five systems and six to ten criteria. The document notes that "not established" means the public source does not mention a feature, not that the feature is missing. It is worth saying this aloud at the defense; the supervisor asked for the same clarification (SF-04).

A5. Technology comparison. Status: done

The technology comparison covers three key decisions (backend platform, data storage and client to server interaction) with three options each, and gives a reasoned choice for every one. The frontend comparison and the authentication and session decisions add two more. This exceeds the requirement of three decisions with two or three alternatives each.

A6. Requirements specification. Status: done

The SRS contains functional requirements FR-01 to FR-17 and non-functional requirements NFR-01 to NFR-06, each with a MoSCoW priority, an acceptance criterion and a verification method. Security and privacy requirements are kept in a separate document and linked through

a traceability matrix. Requirements also appear in requirements.md, but that file states that the SRS is authoritative, so the SRS is the one to show.

A7. Architecture and diagrams. Status: done

The architecture document contains the system context, use case, component and ER diagrams as editable Mermaid code, together with a written description. A data flow diagram is kept in dfd.mmd, a PlantUML use case diagram in usecase.puml, and readable images in the exports folder. Two small issues remain: the use case diagram exists in both Mermaid and PlantUML, so the team should confirm they match, and SSP-41-report.md links to er.mmd and components.mmd, which no longer exist. The supervisor also asked the team to mark which components are already implemented and which are planned (SF-07).

A8. UX and prototype. Status: done

The clickable prototype is a React and TypeScript application with separate workspaces for parents, specialists and administrators, and it builds without errors. Key flows are described in ux-scenarios.md, and the planned API is described in openapi.yaml. However, there is no hosted demo and no screenshots in the repository, and prototype/README.md marks both as deferred. A committee member without Node.js cannot see the prototype. Adding four to six screenshots or publishing the build on GitHub Pages would fix this. The presentation already shows the main screens on slides 6 to 9.

A9. Verification and evaluation plan. Status: done

The verification plan defines test levels, 13 metrics with targets and measurement conditions, a test plan for the OWASP Top 10, and two baselines for comparison: a naive version of the API without the security design, and the OWASP ASVS Level 1 checklist. This is one of the strongest documents in the repository, and it answers the committee question of how the project aim will be demonstrated. The supervisor endorsed the naive baseline (SF-09).

A10. Project management and risks. Status: partial

The project management document assigns responsibilities, sets milestones and lists dependencies D1 to D7 and risks R1 to R10 with likelihood, impact and owners. The weak points are outside this file. The YouTrack board requires a login, there is no submission tag, and the Git workflow is not followed in every case, as described in Part B and in the supervisor's comment SF-10.

A11. Supervisor feedback. Status: done

The supervisor, Senior Lecturer A. K. Makatov, reviewed the repository, the YouTrack project and the presentation on 23 October 2026. He gave 12 comments, and the team recorded a decision, an owner, a deadline and a status for each of them (SF-01 to SF-12, section 7). His conclusion is that analysis and design are at a good level and that the project is recommended for Pre-Defense 1. The presentation now has a dedicated slide (slide 18, "Supervisor

feedback"). The updated supervisor feedback log is prepared as a change to docs/supervisor-feedback.md, together with matching updates to the README, submissions/pre-defense-1.md and project-management.md. It must be merged through a reviewed pull request before the tag is created (section 5).

3. Part B. GitHub repository: the 9 requirements

B1. One permanent repository. Status: met

The project has one repository with a clear name, created on 19 October 2026 and used ever since. It is public, so the supervisor and the committee can open it without extra access.

B2. Individual accounts and commits. Status: partially met

Each member commits from their own account: AknurOE is Aknur Ondasyn, Greatjobson is Beibarys Zhumagali and Luun0 is Yerkebulan Zhuparbay. No shared account is used. Excluding merges, Beibarys has 20 commits, Yerkebulan 12 and Aknur 10. The usernames Greatjobson and Luun0 do not reveal who the students are, so a short table mapping usernames to names in the README would remove any doubt.

B3. Completed README. Status: met

After PR #12 (SSP-45) the README covers the full list: the title and a short description, the problem statement, target users in their own section, the aim and expected outcomes, the scope and current status, the selected technologies with the alternatives considered, the team with roles and the supervisor, links to YouTrack and to the documentation for each of the 11 components, the repository structure, and instructions for running the prototype and the checks. The only remaining suggestion is the username table from B2.

B4. Clear structure and no secrets. Status: met

Code is split into frontend and backend, documentation is in docs, diagrams are in docs/diagrams and a test folder exists at backend/tests. Configuration is covered by .gitignore, .env.example with placeholder values and docker-compose.yml. The secret scan in CI passes. The tests folder is still empty, which is acceptable because the backend has not been implemented yet.

B5. Progressive commits. Status: partially met

The commit history starts on 19 October, and commit messages generally describe a single piece of finished work. The distribution, however, is uneven: 5 commits on 19 October, 4 on 21 October, 4 on 22 October and 29 on 23 October, excluding merges. The committee may read this as the project being uploaded the day before the pre-defense. The team should be ready to explain that earlier work was drafted outside the repository, and from now on should commit smaller pieces as they are finished. The supervisor raised the same issue (SF-10), and the team agreed that every member will commit at least once a week.

B6. YouTrack tasks linked to branches and commits. Status: partially met

Most branches are named after tasks (for example SSP-29-srs and SSP-43-openapi-draft), and 27 of the 42 non-merge commits include an SSP task ID. The 8 frontend commits, commits c8bd92d and 8bf7b30, and pull requests PR #5 and PR #11 have no task ID. It was not possible to check whether YouTrack tasks have an owner, a deadline and acceptance criteria, or whether they link back to pull requests, because the board requires a login.

B7. Branches, pull requests and review. Status: partially met

The team works through branches and pull requests, and 10 of the 12 pull requests were approved by a different team member before merging. There are three exceptions: PR #11 and PR #12 were merged without any review, and commit 8bf7b30 was pushed directly to main. In addition, every review is a plain approval without comments, so there is nothing yet to show for the requirement to present review comments and the changes they led to. The supervisor's review now provides documented comments and decisions, but in-repository review comments are still needed.

B8. Pre-defense tag. Status: not met

The repository has no tags at all. The README and submissions/pre-defense-1.md both refer to a pre-defense-1 tag that does not exist. The tag should be created only after all fixes for this pre-defense, including the supervisor feedback log, are merged, so that it points to the version the team actually presents.

B9. Readiness to demonstrate. Status: partially met

The team can show the history and the contribution of each member, and the prototype can be started with three commands. A complete trace from task to result is ready and is described in section 6. The supervisor's comments and the team's responses can be shown in the feedback log once it is merged. What cannot be shown yet are review comments inside pull requests and the YouTrack side of the trace, unless the committee is given access to the board.

4. Technical checks

The frontend was installed with npm ci, type checked with tsc and built with vite build, and all three steps completed without errors. npm audit reported no vulnerabilities. The latest CI run on main passed, covering go test, govulncheck, npm audit and the gitleaks secret scan. There are no Go test files yet, so go test currently has nothing to check.

The documentation contains 119 Markdown links. All 15 external links open; the Wiley, SAGE and JMIR pages refuse automated requests, but their DOIs are registered. Six internal references are broken or outdated, and they are still present at commit aed3db2:

File	Reference	Problem
SSP-41-report.md	diagrams/er.mmd	The file does not exist; the ER diagram is now in architecture.md
SSP-41-report.md	diagrams/components.mmd	The file does not exist
SSP-41-report.md	beibarys/docs/architecture.md	Old folder structure; now docs/architecture.md
frontend/tech-comparison.md	docs/authentication-and-session-decisions.md	The file moved to docs/decisions/
prototype-security-review.md	docs/prototype/	No such folder; the folder is /prototype
traceability-matrix.md	docs/dpia-summary.md	The file does not exist

There is also one contradiction. submissions/pre-defense-1.md says the survey recruitment method was not recorded, while survey-method.md describes it. Finally, two documents (frontend/tech-comparison.md and evidence-to-requirements.md) are written in Russian while the rest are in English, which is worth confirming with the supervisor.

5. Recommended actions before the defense

The order matters, because the tag must point to the final version.

First, record the supervisor's review. **Done in content:** the review was received on 23 October 2026, and the feedback log (SF-01 to SF-12), the README status row, submissions/pre-defense-1.md and project-management.md have been updated. Open one pull request with a task ID, for example SSP-xx-supervisor-feedback, and ask another team member to review it before merging. Ask the supervisor to confirm the wording of the recorded comments.

Second, make YouTrack visible. Give the supervisor and the committee access, or enable guest access to the project. As a fallback, add a table of tasks with ID, owner, priority, deadline and pull request link to submissions/pre-defense-1.md, together with a screenshot of the board.

Third, open one pull request with a task ID that fixes the six references above, adds a username table to the README, removes the contradiction about recruitment, and adds prototype screenshots to the prototype folder. Ask another team member to review it and leave at least one real comment.

Fourth, leave a review comment on PR #11, PR #12 and commit 8bf7b30 after the fact. Do not rewrite history.

Fifth, once everything is merged, create and push the tag with git tag -a pre-defense-1 -m "Pre-defense 1 submission" followed by git push origin pre-defense-1. After that, never move the tag.

From now on, include the task ID in every commit and pull request, write at least one meaningful comment in every review, and commit work in small pieces as it is completed. These rules are also the team's response to the supervisor's comment SF-10.

6. Preparing the demonstration

For the required trace from task to result, the clearest example is SSP-43. It starts with the task in YouTrack, continues in the branch SSP-43-openapi-draft, goes through PR #10, which AknurOE approved, and ends with the file docs/openapi.yaml. Two further examples are SSP-29, which leads through PR #7 and its six revision commits to docs/srs.md, and SSP-4 with SSP-11, which lead through PR #1 to the threat model and the verification plan. Once merged, the supervisor feedback pull request gives a fourth example that links a review comment to a concrete change.

The committee is likely to ask the following questions.

Question	Suggested answer
Why a survey and not interviews?	The survey was a quick first step to test our assumptions. The sample is small and we state that clearly. Our supervisor asked for the same, so three to five interviews are planned before the next pre-defense.
Where is the backend?	It is not required at this stage. We have Go domain models, an OpenAPI contract and an ER model. Implementation is the next milestone, starting with Access and Profiles, as the supervisor recommended.
How is your project different from existing systems?	Based on public descriptions, none of the four systems combines visibility control for individual entries, access revocation and auditing. That combination is the engineering focus of the project.
How will you prove that access control works?	With a test for every cell of the access matrix, and by running the same tests against a naive baseline to show which ones it fails.
Did your supervisor review the work, and what changed?	Yes, on 23 October. He gave 12 comments and we accepted all of them. For example, after his comment on the Git process we agreed on weekly commits and mandatory peer review, and after his comment on children's data we added legal-representative consent and data localisation to our legal checklist. Each comment is linked to a change in the feedback log.
Who did what?	Aknur worked on security, privacy requirements and the verification plan. Beibarys worked on the proposal, SRS, architecture, OpenAPI contract and CI. Yerkebulan worked on user research, the prototype and UX. All of this is visible in the pull requests.

7. Supervisor feedback

Reviewer: Senior Lecturer Arstan Kutzhanovich Makatov. Date: 23 October 2026. Materials reviewed: the GitHub repository (README, docs, frontend, backend), the YouTrack project SSP and the Pre-Defense 1 presentation.

7.1 Overall assessment

According to the supervisor, the topic is socially significant and has enough engineering depth. The problem is clearly defined and the MVP scope is realistic for a diploma project. The focus on access control, privacy and auditing gives the project sufficient engineering complexity. The analysis and design stage is at a good level: the proposal, the SRS with acceptance criteria, the traceability matrix, the comparisons of existing systems and technologies, the diagrams, the threat model, the verification plan with a baseline, and a clickable prototype in three languages are in place. Roles are clearly divided, and each member has their own commits and pull requests.

7.2 Comments and the team's decisions

ID	Area	Supervisor's comment	Decision and change	Owner / status
SF-01	Scope	Scope is realistic. Keep the MVP to access and roles, profiles and links, sessions, a visibility-controlled diary and audit. Exclude video calls, billing and integrations.	Accepted. Out-of-scope list confirmed in the proposal and README.	Beibarys / done
SF-02	Evidence	Seven survey answers are exploratory, not proof. Conduct 3–5 semi-structured interviews. Keep single-source requirements marked as hypotheses.	Accepted. Interviews with 2 parents and 2–3 specialists planned; hypothesis labels kept.	Yerkebulan, Aknur / before PD-2
SF-03	Literature	14 sources are a sufficient base. Ensure at least 8 peer-reviewed sources, add 2–3 on RBAC/ABAC for personal or health data, use one citation style.	Accepted. Literature review to be extended and reformatted.	Aknur / planned
SF-04	Existing systems	Each "?" cell must cite a source or be stated as "not found in public sources".	Accepted. Legend and sources clarified in the comparison.	Yerkebulan / in progress
SF-05	Technologies	React + TypeScript, Go and PostgreSQL are justified. Decide hosting with Kazakhstan data localisation in mind before deployment.	Accepted. Hosting added as an open item in the legal checklist.	Beibarys, Aknur / planned
SF-06	Requirements	Every Must requirement needs a measurable acceptance criterion and a linked test in the traceability matrix.	Accepted. Must requirements reviewed against the matrix.	Aknur / in progress

ID	Area	Supervisor's comment	Decision and change	Owner / status
SF-07	Architecture	Diagrams are consistent, but only the client is built and the backend domain is not connected to the UI. Mark built vs planned components; start transport/http from openapi.yaml.	Accepted. Component diagram to distinguish built and planned parts; backend starts from the OpenAPI contract.	Beibarys / stage 2
SF-08	Prototype	The prototype shows the key scenarios of all three roles. Run a usability test with 5 users: task success, time on task, SUS.	Accepted. Usability test scheduled after the interviews.	Yerkebulan / before PD-2
SF-09	Verification	Keep the naive baseline. The pilot-size assumption is accepted but must be labelled. RBAC/IDOR tests should come first in CI.	Accepted. Assumption labelled in the verification plan; RBAC/IDOR tests first in stage 2.	Aknur / done (plan)
SF-10	Git process	Most commits were made on 22–23 October, and PRs #11 and #12 were merged without review. This does not show progressive work.	Accepted. Weekly commits per member, peer review for every PR, task ID in branch and PR, PR link in YouTrack, tag pre-defense-1.	All / in progress
SF-11	Child data	The diary may contain health information. Check the RK Law "On Personal Data and Their Protection" (legal-representative consent, retention, localisation), consult the university lawyer, use synthetic data only.	Accepted. Items added to the legal checklist; synthetic-data rule kept.	Aknur / planned
SF-12	Next stage	Start development with Access and Profiles (sign-up and login, tokens, RBAC, parent–child–specialist links) with automated access tests.	Accepted. Milestone order confirmed in the project management document.	Beibarys, Aknur / stage 2

7.3 Impact on this review

The supervisor's comments match the main findings of this review: the limited user evidence (A2), the uneven commit history and missing reviews (B5, B7), and the missing tag (B8). Component A11 is now done. The comments also add three points that this review did not

cover: the legal requirements for children's data, the hosting decision, and the need to separate implemented and planned components in the architecture.

7.4 Conclusion of the supervisor

> *"The materials meet the Pre-Defense 1 requirements for the analysis and design stage. The project is recommended for Pre-Defense 1. Key tasks before Pre-Defense 2: implement the backend (Access and Profiles), collect additional user evidence, and maintain a regular commit history with reviewed pull requests." — A. K. Makatov*